Project Proposal: Modular 6-DOF Industrial Robotic Arm

NEURAL NEXUS

- One arm. Unlimited applications -

1. Executive Summary

This proposal outlines the strategic development and rapid market entry of an affordable, industrial-grade 6-Degrees of Freedom (DOF) robotic arm by **NEURAL NEXUS**. Industrial automation in Sri Lanka is currently stifled by the prohibitive cost of imported systems (1.0–1.5 million LKR). Our mission is to democratize high-end robotics by providing a performance-equivalent, modular system priced under 0.8 million LKR. This system is specifically tailored for the unique constraints of local SMEs, garment factories, and educational institutions, facilitating a rapid transition towards “Industry 4.0” through localized innovation and a compressed 3-month development cycle.

2. Market Justification & Target Segments

By 2026, the global shift toward **“Physical AI”** has transformed automation from a luxury into a competitive necessity. While developed nations have already reached a peak of efficiency through the implementation of **“dark factories”**—fully autonomous, unlit production lines emerging markets like Sri Lanka face a significant innovation-execution gap.

Despite a surplus of innovative ideas, the lack of local resources and high import costs typically delay the domestic adoption of such technologies by an estimated **6 to 8 years**. Our development of a localized robotic arm serves as a strategic intervention to bridge this gap through three primary objectives:

- **Lead-Time Reduction:** By providing immediate, locally-available automation, we significantly reduce the lead time for industries that are currently underserved by global manufacturers focused on high-volume automotive lines.
- **Market Alignment:** We are entering a market where the industry explicitly expects local automation solutions. By offering a high-precision, low-cost alternative, we make advanced robotics accessible to a broader range of local enterprises.
- **Ecosystem Development:** Beyond the hardware, this project acts as a catalyst for a domestic supply chain. Our approach encourages the production of **custom motors** and specialized **industrial materials** within Sri Lanka, fostering a self-sustaining ecosystem for local high-tech manufacturing.

Primary Target Segments

- **SME Manufacturing:** Packaging, food processing, bottling, and repetitive material handling.
- **Garment Industry:** Automating fabric sorting, folding, and repetitive assembly tasks.
- **Electronics Assembly:** High-speed PCB placement, soldering, and quality inspection.
- **Educational Institutes:** Providing universities with a robust R&D platform for AI training.STEM Education
- **Custom Automation:** Local startups requiring non-standard tool integrations for niche processes.

3. Technical Specifications

To ensure industrial-grade reliability, our system adheres to the following engineering benchmarks:

Component/Feature	Specification / Requirement
Degrees of Freedom (DOF)	6 (Waist, Shoulder, Elbow, and 3-Axis Spherical Wrist)
Rated Payload / Reach	3.0 kg (Rated) / 750 mm (Spherical workspace)
Repeatability Spec	± 0.5 mm (High-resolution absolute encoders)
Structural Integrity	CNC-machined Aerospace-grade Aluminum and Steel
Control Architecture	Closed-loop real-time control; Inverse Kinematics (IK) processing
Drive System	High-torque stepper motors with customized plantery gearboxes
Safety Systems	Emergency Stop, Overcurrent Protection, Physical limit Sensors

Table 1: Core Technical Benchmarks for Industrial Deployment

4. Modular Ecosystem: Swappable Tool-Heads

The versatility of the **NEURAL NEXUS** arm lies in its rapid-change modular interface:

- **Gripper Head:** Mechanical picking and sorting for small parts.
- **Vacuum Suction Head:** Ideal for box handling, palletizing, and delicate flat sheets.
- **Screwdriver Head:** Precision torque-controlled assembly for electronics.
- **Vision Module:** Integrated AI-enabled camera for real-time object recognition.

5. Market Approach & Sales Strategy

NEURAL NEXUS differentiates itself through a disruptive price-to-performance ratio and a localized support ecosystem designed to minimize industrial downtime.

Competitive Pricing Model

Our pricing is structured to lower the barrier to entry while maintaining sustainable margins.

- **Target Selling Price (Base Unit): 350,000 – 500,000 LKR**.
- **Price Advantage:** $\sim$50% lower than entry-level imported units from major global brands.
- **Add-on Modules:** Specialized heads priced at 35,000 – 85,000 LKR.
- **Software Tiers:** Core control is included; AI-vision upgrades are sold as yearly licenses.

Sales Channels & Market Penetration

To capture the Sri Lankan market effectively, we utilize a multi-pronged approach:

- **The "Trojan Horse" Strategy:** Enter a factory by automating one specific bottleneck (e.g., bottling) to demonstrate ROI within 6 months, justifying further installations.
- **Automation-as-a-Service (AaaS):** Offer a lease model where clients pay per 1,000 cycles.
- **B2B Partnerships:** Partnering with local consultants who design production lines.
- **Education Bundles:** Providing technical colleges with a "Robotics Lab Starter Kit."

The Local Support Advantage

The primary deterrent for local firms is the lack of on-site support for international systems.

- **24/48-Hour Service Guarantee:** On-site support and local inventory of spare parts.
- **Customization Service:** Bespoke tool-head design for localized industry needs (e.g., textile handling).

6. Project Budget (Estimated for MVP Prototype)

Item Category	Description	Cost (LKR)
Actuators & Feedback	6x Industrial Grade High-torque steppers with Encoders	150,000
Controller & Logic	Industrial-grade Central Controller and I/O interface	45,000
Structural Components	CNC-machined metal frame, precision bearings, and housing	50,000
Initial Tooling	Prototypes for Gripper and Vacuum Suction modules	35,000
Sensors & Safety	Limit switches, Emergency stop, Power management circuits	40,000
R&D & Iteration	Software development and rapid prototyping contingency	60,000
TOTAL COST		**380,000 LKR**

Table 2: Prototype Development Budget (MVP Phase)

7. Accelerated Implementation Timeline (3 Months)

The project utilizes a parallel-track development strategy to meet a 12-week deployment goal.

- **Month 1:** Kinematic modeling, structural design validation, and component sourcing.
- **Month 2:** CNC fabrication of frames, actuator mounting, and control logic integration.
- **Month 3:** Stress testing, UI/UX refinement for friendly visible control, and pilot deployment.

8. Future Improvements

NEURAL NEXUS aims to integrate **Agentic AI** for autonomous task re-planning.A core improvement of this project is the development of a highly intuitive Graphical User Interface (GUI) designed to bridge the technical gap between complex robotics and the end-user. Traditionally, reconfiguring a robotic arm for a new task requires specialized knowledge of programming languages or proprietary industrial controllers. Our solution removes this barrier